Planar graphs, obviously, have a crossing number equal to 0; both non-planar graphs we have met earlier in the chapter, K_5 and $K_{3,3}$, have a crossing number equal to 1.

15.3.1 Finding the Crossing Number

Kuratowski's theorem tells us what's the necessary and sufficient condition for a graph to be planar, but it doesn't help much with computing the minimum crossing number of non-planar graphs. The problem of finding the crossing number for a non-planar graph has been investigated far less than planarity testing/embedding: while there are several efficient algorithms that provide planar embeddings for planar graphs, at date there isn't an efficient algorithm that can find the minimum crossing number for generic graphs.

As a matter of fact, it has been proven that determining the crossing number of a generic graphs is an *NP-complete* problem.

There are, however, notable exceptions[23] if we narrow the field; for example, it has been recently proven[24] that there is a simple algorithm to check if a non-planar graph has crossing number 1:

Assuming that `G` is a non-planar graph (and hence it has at least 5 vertices, as a consequence of Kuratowski's theorem), for each pair of non-adjacent[25] edges a→b, c→d, we remove both edges, and add a new vertex v, and 4 new edges a→v, v→b, c→v, v→d. If the new graph obtained is planar, then the crossing number of the original graph is exactly 1.